

ELECENG 2EI4

Project 2: Voltage Controlled Switch

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Switch Properties

Ideal Switch properties:

A switch is an electrical component that can disconnect or connect the conducting path in an electrical circuit. This usually involves interrupting the electric current or diverting it from one conductor to another. There are several ideal properties of a switch regarding resistance, current, and voltage properties in its ON/OFF states. The following properties are listed in the table below.

Property	Ideal Behaviour	Mathematical Expression
ON Resistance	Zero	$R_{ON} = 0$
OFF Resistance	Infinite	$R_{OFF} = \infty$
Bidirectional conduction	Current can pass in either direction	$-\infty \leq I \leq \infty$
Voltage and Current Rating	No rating on either	$V_{max} = I_{max} = +\infty$ $V_{min} = I_{min} = -\infty$
Voltage Drop (ON)	Zero	$V_{drop} = 0$
Current (OFF)	Zero	$I = 0$
Switching Time	Instantaneous	$t_r = t_f = 0$
Power Dissipation	Zero	$P = 0$
Bandwidth	Infinite	$f = \infty$
Leakage	None	$I_{leak} = 0$

Non-Idealistic (Real) properties of switches:

Real switches deviate from their ideal properties due to physical device limitations such as finite resistances, load capacitances, switching speeds, and charge storing effects. These external effects cause major deviations in a MOSFET-based switch behaviour. The following table summarizes these non-idealistic behaviours.

Non-Ideality	Physical Cause	Quantitative Expression	Qualitative Effect
Limited Current Rating	Real switches only carry a finite current. Exceeding this limit causes overheating or permanent damage	Device-specific rating (eg. 1-5A or 5-30A)	Limits maximum load current and cause a thermal increase.
Operational Voltage Range	Switches having a maximum voltage range. Exceeding it may cause breakdown or clipping of the signal. Output voltage is slightly less than input	$V_{out} \leq V_{supply} - V_{sat}$	Limit maximum signal amplitude and affects accuracy of output.
Finite Switches Speed	Switching is not instantaneous and there exists a delay between rise and fall times due to load capacitances.	$t_{ON} = t_d + t_r$ $t_{OFF} = t_s + t_f$ $t_r \approx 2.2R_{ON}C_L$ $t_f \approx 2.2R_{OFF}C_L$	Causes propagation delay to other components and limits higher speed operations
Non-Zero ON Resistance	Close switch has a finite resistance, causing voltage drops and power loss	$V_{drop} = I \times R_{ON}$ $P_{abs} = I^2 \times R_{ON}$	Reduced output voltage and heat dissipation.
Finite OFF Resistance	Small currents flow even when the switch is open due to imperfect insulation	$I_{leak} = \frac{V_{drop}}{R_{OFF}}$	Causes a loss in input current.
Non-Ideal Bidirectionality	Real switches may pass less efficiently in one direction than the other	$V_{AB} = \begin{cases} I \times R_{ON,f} & \text{if } I > 0 \\ I \times R_{ON,r} & \text{if } I < 0 \end{cases}$	Forward and reverse voltage drops may slightly differ.
Bandwidth specific	Internal capacitances cause certain frequencies to get attenuated.	$f_{min} = \frac{1}{2\pi R_{ON}C_{interanl}}$	High frequency signals get reduced with lower and equal frequencies (f_{min}) remain unaffected.
Charge injection	Switching between ON and OFF states cause charges stored in the MOSFET to be injected in the output.	$\Delta V = \frac{Q_{inj}}{C_{load}}$	Causes small voltage spikes in the output signal.

Test Plan

For the following designs, a test plan will be constructed to measure the correct for the errors associated with the use of non-ideal switches. The main non-ideal properties (from the list above) that will be analysed are the ON and OFF state resistance properties, operational voltage ranges, and bidirectionality properties. All voltages in the circuit are constrained to $V \leq 5V$, as required by the design constraints.

For **ALL** following testing experiments $V_{supply} = 5V$, $V_{GND} = 0V$, R_L is known and is connected from $V_2/V_A/V_B$ to $V_{GND} = 0V$

Switch Type 1 – Test Plan

Test 1: ON-State Resistance Measurement

When the MOSFET switch is ON, there will always exist a small resistance that will affect the output voltage. Voltage division rules and Ohm's law can be applied to calculate this resistance as well as the power loss by the switch.

Set:

- $V_{control} = 0V$ (ON)
- $V_1 = 5V$

Measure/Observe:

- V_2

Calculate/Determine:

- $V_{drop} = V_1 - V_2$
- $R_{ON} = R_L \left(\frac{V_1}{V_2} - 1 \right)$
- $P_{abs} = I^2 \times R_{ON}$

Test 2: OFF-State Leakage Current

When the MOSFET switch is OFF, there will always exist some small current that will leak into the output circuit. This is due to the impossibility of infinite resistance when OFF. The leakage current can be calculated via Ohm's law.

Set:

- $V_{control} = 5V$ (OFF)
- $V_1 = 5V$

Measure/Observe:

- V_2

Calculate/Determine:

- $I_{leak} = \frac{V_2}{R_L}$

Test 3: Voltage Operating Range

When the MOSFET switch is ON there may be an operational voltage range for the input signal, everything beyond that range leading to extreme non-ideal behaviours. Assuming that the supply voltage stays at 5V, the input voltage can be altered to determine the minimum and maximum input voltage ranges where the switch behaves the most ideal.

Set:

- $V_{control} = 0V$ (ON)
- $V_1 = 0V - 5V$ (Ramp function)

Measure/Observe:

- Output waveform integrity
- Distortion or abnormal behaviour

Calculate/Determine:

- Minimum operating voltage (V_{min})
- Maximum operating voltage (V_{max})

Test 4: Bidirectional Conduction

When the MOSFET switch is ON there may be deviations to its behaviour depending on the direction the current is passed through. This deviation is mainly due to slight variations in the ON resistance of the switch circuit. Hence the ON resistance can be measured when switching polarities and compared.

Set:

- $V_{control} = 0V$ (ON)
- $V_1 \geq V_2$ and $V_1 \leq V_2$ (Reversing polarities)

Measure/Observe:

- V_1
- V_2

Calculate/Determine:

- $R_{ON,f}$ and $R_{ON,r}$, Using the formula: $R_{ON} = R_L \left(\frac{V_1}{V_2} - 1 \right)$
- Compare if: $R_{ON,f} \approx R_{ON,r}$

Switch Type 2 – Test Plan

Test 1: ON-State Resistance Measurement (Active Output Path)

When the selected path is ON, a small ON resistance will exist between V_1 and the active output (either V_A or V_B). This resistance will cause a voltage drop and power loss.

Case 1: V_A Active

Set:

- $V_{control} = 0V$ (A active)
- $V_1 = 5V$

Measure/Observe:

- V_A

Calculate/Determine:

- $V_{drop} = V_1 - V_A$
- $R_{ON,A} = R_L \left(\frac{V_1}{V_A} - 1 \right)$
- $P_{abs} = I^2 \times R_{ON}$

Case 2: V_B Active

Set:

- $V_{control} = 5V$ (B active)
- Same conditions as Case 1

Measure/Observe:

- V_B

Calculate/Determine:

- Same calculations as Case 1, but with V_B instead of V_A

Final Comparison:

- Compare if $R_{ON,A} \approx R_{ON,B}$
- If significantly different, switch has asymmetric properties.

Test 2: OFF-State Leakage Current

When one output is selected, the other output should ideally be completely isolated.

Case 1: V_A Active ($V_{control} = 0V$)

Set:

- $V_{control} = 0V$ (A active)

- $V_1 = 5V$

Measure/Observe:

- V_B

Calculate/Determine:

- $I_{leak,B} = \frac{V_B}{R_L}$

Case 2: V_B Active ($V_{control} = 5V$)

Repeat steps as Case 1, but with measuring V_A

Test 3: Voltage Operating Range

When the switch is ON, the active output should track the input voltage linearly within the allowed range.

Set:

- $V_{control} = 0V$ (or $5V$ for opposite case)
- $V_1 = 0V - 5V$ (Ramp function)

Measure/Observe:

- Output waveform integrity (V_A or V_B)
- Distortion or abnormal behaviour

Calculate/Determine:

- Minimum operating voltage (V_{min})
- Maximum operating voltage (V_{max})

Test 4: Bidirectional Conduction

The conduction symmetry of each path should be verified since MOSFETs may behave differently depending on current direction.

Case 1: V_A Active

Set:

- $V_{control} = 0V$ (A active)
- $V_1 \geq V_A$ and $V_1 \leq V_A$ (Reversing polarities)

Measure/Observe:

- V_1
- V_A

Calculate/Determine:

- $R_{ON,A,f}$ and $R_{ON,A,r}$, Using the formula: $R_{ON} = R_L \left(\frac{V_1}{V_2} - 1 \right)$
- Compare if: $R_{ON,A,f} \approx R_{ON,A,r}$

Case 2: V_B Active

Repeat Case 1 steps but with B active ($V_{control} = 5V$) and measure V_A instead of V_B and calculate $R_{ON,B,f}$ and $R_{ON,B,r}$.

Switch 1

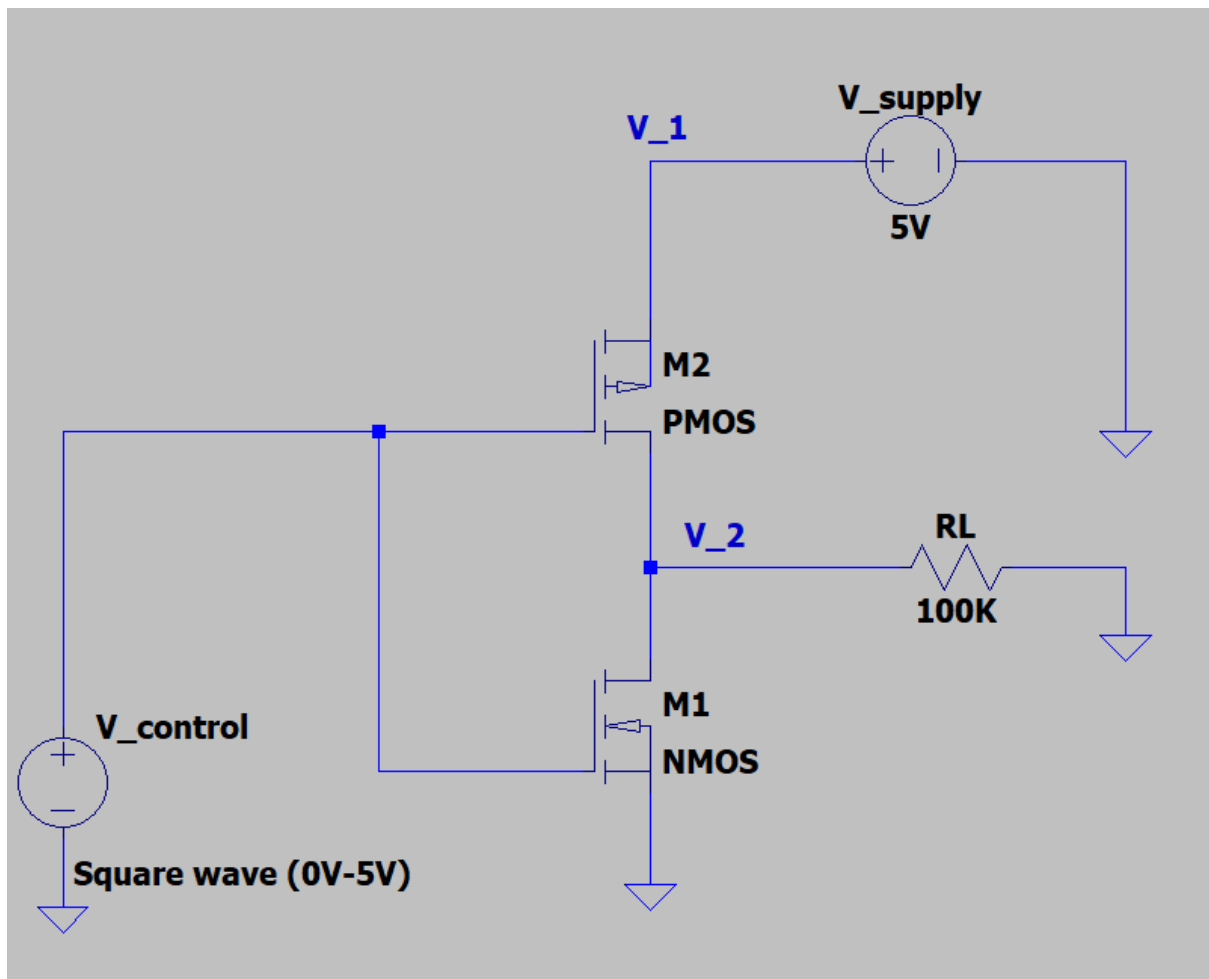


Figure 1: LSpice Schematic of Circuit

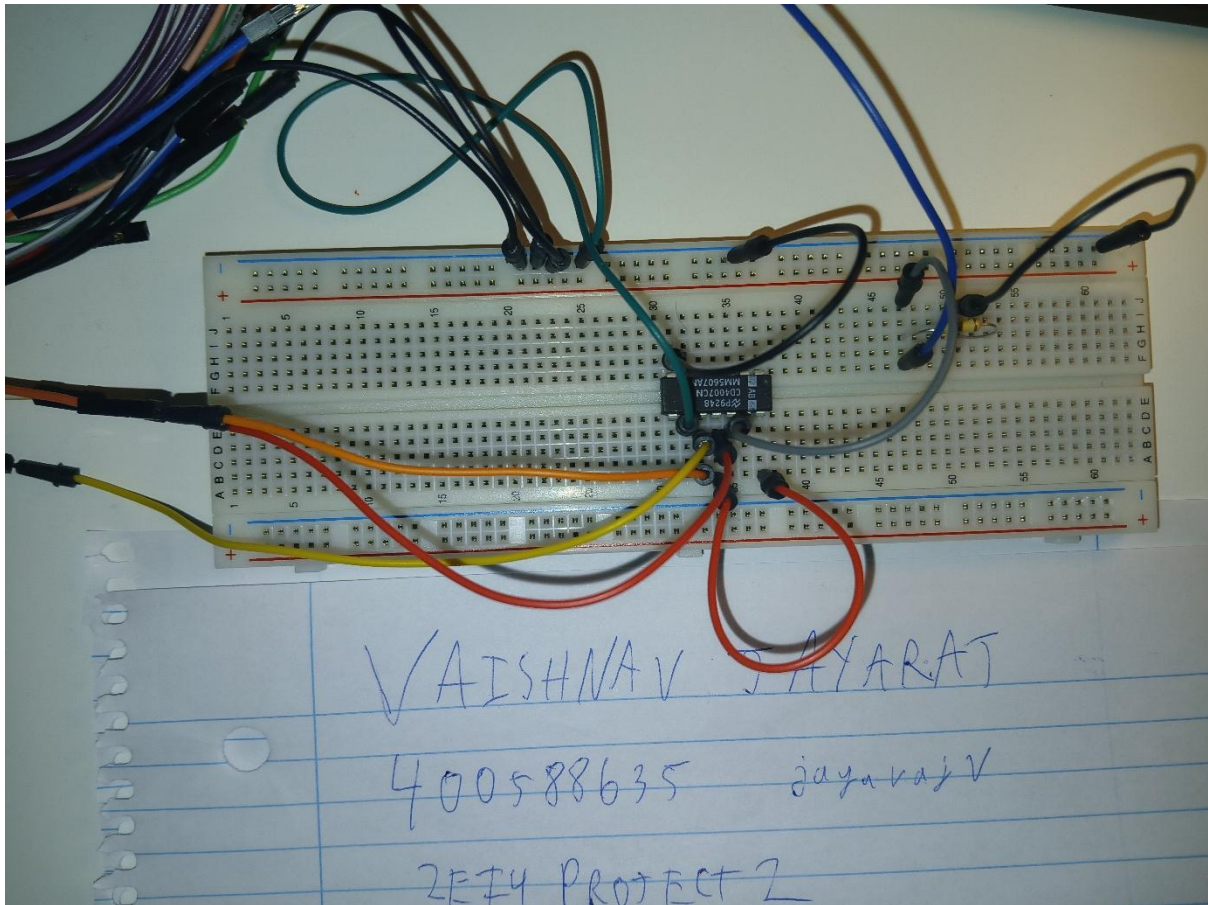


Figure 2: Physical construction of circuit

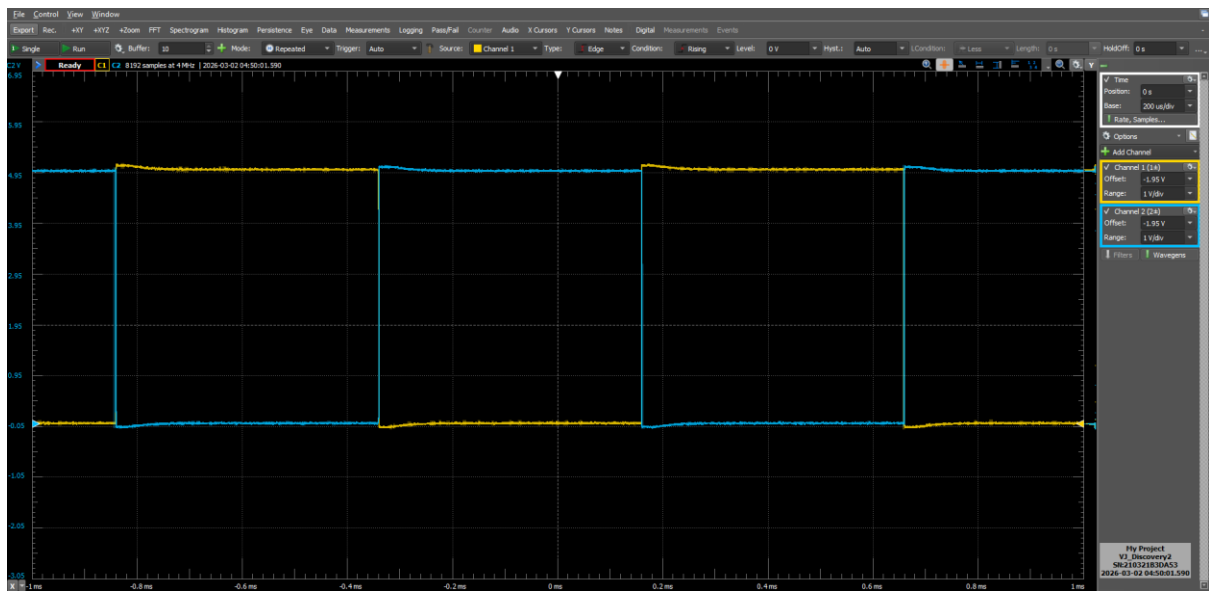


Figure 3: Scope Reading of V1 and V2

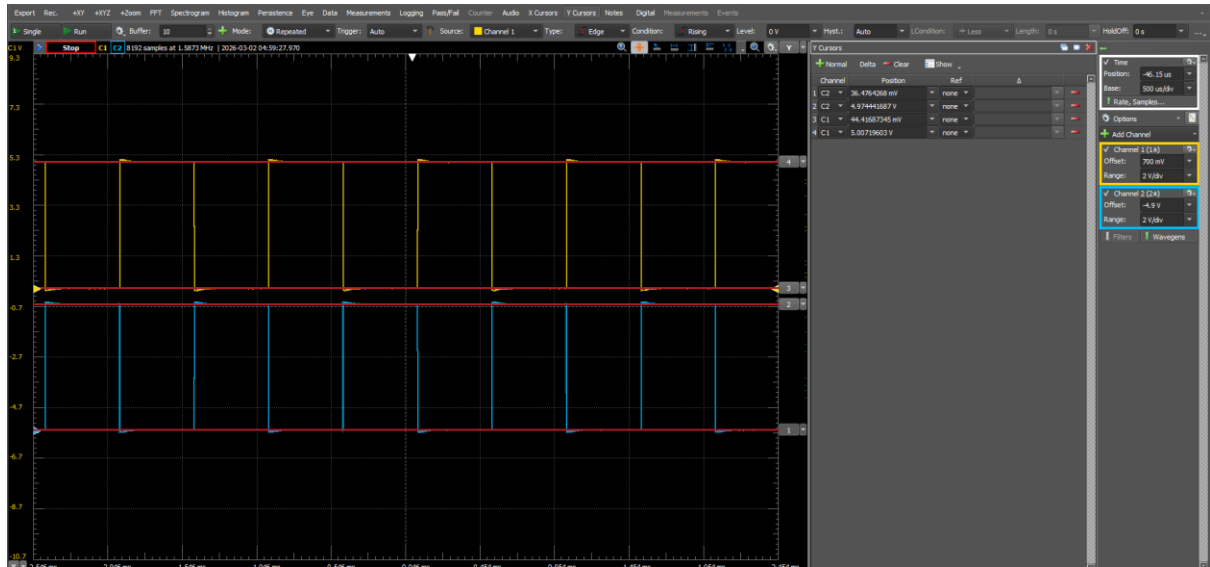


Figure 4: Max amplitudes of V1 and V2

Variable	Measured result
V_1	5V
$V_{2,ON}$	4.97444V
$V_{2,OFF}$	36.476 mV
V_{drop}	25.568 mV
R_{ON}	513.8 Ω
P_{abs}	1.27 μ W
I_{leak}	364.76 nA
V_{min}	0V
V_{max}	3.5V – 4.0V
$R_{ON,f}$	513.8 Ω
$R_{ON,r}$	535.2 Ω

Figure 5: Table of Measured results

Theoretical Calculations:

From the datasheet for the MOSFET used (CD4007B IC)

Parameter	NMOS	PMOS
Threshold voltage V_T	+2.0 V	–1.5 V
Process transconductance K_p	111 μ A/V ²	55 μ A/V ²

1) Theoretical ON-State Resistance R_{ON}

When the MOSFET is ON, it operates in the linear region. Hence:

$$I_D \approx k(V_{GS} - V_T)V_{DS}$$

$$\therefore R_{ON} = \frac{V_{DS}}{I_D} = \frac{1}{k(V_{GS} - V_T)}$$

NMOS ON Resistance

Assume $V_{GS} = 5$ V:

$$V_{GS} - V_{T1} = 5 - 2.0 = 3.0 \text{ V}$$

$$R_{ON,n} = \frac{1}{111 \times 10^{-6} \times 3.0} = \frac{1}{333 \times 10^{-6}} \approx 3.0 \text{ k}\Omega$$

PMOS ON Resistance

Assume $V_{SG} = 5 \text{ V}$:

$$V_{SG} - |V_{Tp}| = 5 - 1.5 = 3.5 \text{ V}$$

$$R_{ON,p} = \frac{1}{55 \times 10^{-6} \times 3.5} = \frac{1}{192.5 \times 10^{-6}} \approx 5.2 \text{ k}\Omega$$

Total ON Resistance

$$R_{ON,total} = \left(\frac{1}{R_{ON,n}} + \frac{1}{R_{ON,p}} \right)^{-1} = \left(\frac{1}{3000} + \frac{1}{5200} \right)^{-1} \approx 1.9 \text{ k}\Omega$$

2) Theoretical OFF-State Leakage Current

In OFF state, the MOSFET is in cutoff since:

$$V_{GS} < V_T$$

So theoretically

$$I_{leak} = 0$$

Explanation of results:

From the experiments done it can be seen from the graphs and data collected that the circuit designed behaves appropriately like an ideal circuit switch. At the ON state the circuit was measured to have an internal ON resistance of approximately 514Ω . This is extremely low when compared to the theoretical resistance calculation of $1.9\text{K}\Omega$. This large disparity is most likely due to variations in the k value due to the amount of current passed and environmental conditions. Furthermore, it was also visually noted that the MOSFET switch did operate bidirectionally with only a small recorded voltage drop and change in resistance $\Delta R = 21.4\Omega$. When the MOSFET switch was OFF a leakage current of 364.76nA was recorded. This extremely small current is most likely due to static charges and other external factors in the circuit as well as defects in the IC chip. Lastly the best physically possible operational voltage range for the circuit was from 0V to around $3.5\text{-}4.0\text{V}$. Any voltage after this value showed an increase in the voltage drop across the switch and abnormal ON behaviours.

Design Trade-off:

For this design 2 MOSFETs were chosen instead of a single transistor to improve performance. The main performance improvement is bidirectional conduction and reduced maximum voltage drop over the switch. However, this increase in complexity will increase the cost of the circuit as well as introducing more sources of error. Additionally, lowering the ON resistance will require more components, which will increase parasitic capacitances and reduce switching speed.

References:

[1] ADI Analog Dialogue, “ADALM2000 Activity: Build CMOS Logic Functions Using the CD4007 Array” [Online]. Available: <https://www.analog.com/en/resources/analog-dialogue/studentzone/studentzone-september-2022.html> Accessed Mar 1. 2026